

License Plate Detection

Session 2

1. GOAL

- Explore the distribution and properties of a DataSet
- Identify the learnable parameters of an ML method

2. MATERIALS:

Slides: Session slides 2.Challenge1_Detection_Traditional_Approach.pdf

Code: DataExploration.py, LicensePlateDetector.py

Data: real_data.zip, your own data

3. SESSION EXERCICES

Exercise 1. Image Properties Distribution. Explore the distribution of image properties (color, illuminance, saturation) and camera viewpoint (orientation and distance to object) using the script DataExploration.py.

- a) How are color and illuminance computed? What colors have the cars of our data set? Are any colors missing?
- b) How are viewpoint properties computed from the plate geometric properties using cv2.minAreaRect. What codifies the variable plateAngle?
- c) Analyze the histograms and boxplots for the real_set visualized in DataExploration.py. What properties have a different Distribution for the lateral and frontal views? Why?
- d) Compare distribution of Lateral and Frontal views (real_set and new images). using histograms and boxplots. Identify differences in image properties across views and sets. Do both views in the set real_set have same image properties? Is the viewpoint of your Frontal/Lateral data comparable to the distribution of the set real_set? Do your images follow the same the distribution of the real_set?

Exercise 2. Acquisition Protocol and Dataset Completion. From the conclusion of Exercise1, define an acquisition protocol to complete your data set so that image properties have similar distribution for all sets and camera viewpoints are uniform across frontal and lateral views of the different sets. Use this protocol to incorporate another set of images violating the acquisition protocol.

Exercise 3. ML method for Plate Detection. Identification of the Learnable Parameters

- a) Identify the main steps of detectPlates() function..
- b) Analyse sequence and purpose: blackhat → gradient → blur → threshold → morphology
- c) Explore parameters: kernel size, threshold, area, aspect ratio, iterations.

- d) Predict effects of modifications: what happens if steps or values are changed?
- e) Compare results: test with images acquired under protocol and those without it. Can you find parameters that work well for both?